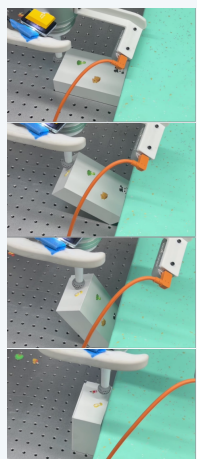


(a) Task-conditioned future learning

Task demonstrations



Training input

Current image



Current wrench



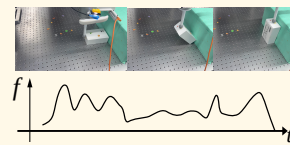
Instruction

Flip the box.

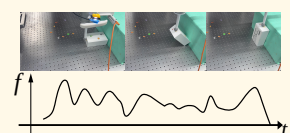
ForceWM
Task-conditioned
world model

Predicted future

Visual-wrench future



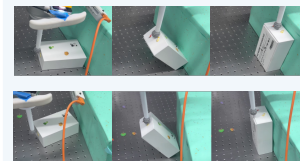
Recorded future



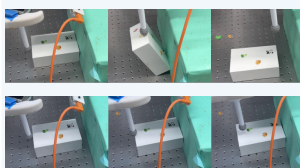
L Supervision

(b) Inverse dynamics from task and play

Task trajectories

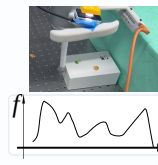


Play trajectories

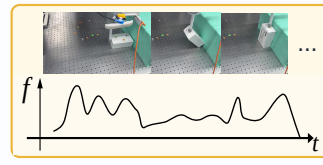


Same recorded window

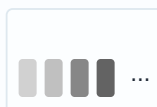
Current observation



Recorded visual-wrench future



Recorded actions



Causal IDM

Inverse dynamics model

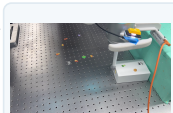
Predicted actions



L Supervision

(c) Online composition

Current image + wrench



Instruction

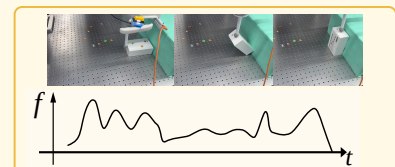
Flip the box.

ForceWM
Task-conditioned
world model

Slow reference refresh

Generated visual-wrench future

Cached reference

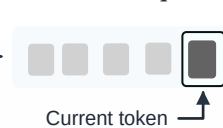


Causal IDM

Inverse dynamics model

Fast action updates

Action sequence



Current token

Pose + gripper
command

Measured wrench feedback



Robot execution

